

Quick Activity 2: Canteen Group Order

A canteen sells rice meals at ₱85 each and drinks at ₱25 each. Every order receives a 10% discount on the rice-meal subtotal only. The canteen also charges a ₱5 packaging fee per rice meal.

INPUT	PROCESS	OUTPUT
Number of rice meals Number of drinks Customer's payment	Calculate the rice-meal subtotal: Rice meals x ₱85 Calculate the drink subtotal: Drinks x ₱25 Calculate the 10% discount on the rice-meal subtotal only: Meal subtotal x ₱5 Calculate the final bill: Meal subtotal - Discount + Drink subtotal + packaging fee Calculate the change: Payment - Final bill	Meal subtotal Drink subtotal Discount Packaging Final bill Change

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START

INPUT riceMeals

INPUT drinks

INPUT payment

mealSubtotal = riceMeals x 85

drinkSubtotal = drinks x 25

discount = mealSubtotal x 0.10

packagingFee = riceMeals x 5

discountedMeals = mealSubtotal - discount

finalBill = discountedMeals + drinkSubtotal + packagingFee

change = payment - finalBill

DISPLAY mealSubtotal

DISPLAY drinkSubtotal
DISPLAY discount
DISPLAY packagingFee
DISPLAY finalBill
DISPLAY change
END

DRY-RUN CALCULATION: QUICK ACTIVITY TWO

CASE TEST 1:

Input:

- Rice meals = 3
- Drinks = 4
- Payment = ₱500.00

STEP	VARIABLE	CALCULATION	RESULT
1	mealSubtotal	3 x ₱85	₱255
2	drinkSubtotal	4 x ₱25	₱100
3	discount	₱255 x 10%	₱25
4	discountedMealSubtotal	₱255 - ₱25.50	₱229.50
5	packagingFee	3 x ₱5	₱15
6	finalBill	₱229.50 + ₱100 +15.00	₱344.50
7	change	₱500 - ₱344.50	₱155

CASE TEST 2:

Input:

- Rice meals = 5
- Drinks = 3
- Payment = ₱600.00

STEP	VARIABLE	CALCULATION	RESULT
1	mealSubtotal	5 x ₱85	₱425
2	drinkSubtotal	3 x ₱25	₱75
3	discount	₱425 x 10%	₱42.50
4	discountedMealSubtotal	₱425 - ₱42.50	₱382.50
5	packagingFee	5 x ₱5	₱25
6	finalBill	₱382.50 + ₱75 + ₱25	₱482.50
7	change	₱600 - ₱482.50	₱117.50

Explain: Why would applying the 10% discount to the entire bill produce an incorrect result?
- The 10% discount applies only to the rice-meal subtotal. Drinks and packaging fees are explicitly excluded from the discount.

For example, in Test Case 1:

Correct discount = 10% of ₱255.00 = ₱25.50

If the entire bill before discount were discounted, the calculation would incorrectly include the ₱100.00 drinks and ₱15.00 packaging fee.

That would give a discount of $10\% \times (\text{₱}255.00 + \text{₱}100.00 + \text{₱}15.00) = \text{₱}37.00$

This is ₱11.50 more discount than allowed. Therefore, applying the discount to the entire bill would produce an incorrect final bill and change the amount.